

Brachytherapy for prostate cancer

Radiotherapy for prostate cancer is sometimes given from inside the body. This is called internal radiotherapy or brachytherapy.

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What is brachytherapy for prostate cancer?

Brachytherapy (internal radiotherapy) gives high doses of radiation directly to the prostate. Areas close by get a much lower dose – for example, the [rectum and the bladder](#).

Brachytherapy can be given:

- with hormonal therapy, or on its own to treat low risk prostate cancer
- before or after external beam radiotherapy for intermediate risk and high risk [early prostate cancer](#) or [locally advanced prostate cancer](#).

There are 2 ways of giving brachytherapy:

- **Radioactive seed brachytherapy or low dose rate (LDR) brachytherapy** - small, radioactive metal 'seeds' are placed in the prostate and stay there permanently. They slowly release radiation until it has been used up. After 9 months, most of the radiation will have gone.
- **High dose rate (HDR) brachytherapy** - thin tubes are placed in the prostate. They are attached to a machine that sends a radioactive source into the tubes for a set time. The source and tubes are then removed.

Doctors may use high dose rate (HDR) brachytherapy with external radiotherapy to treat [locally advanced prostate cancer](#). They sometimes call it a boost dose. You usually have it 1 to 2 weeks before you start external radiotherapy or when you finish external radiotherapy.

There may be some differences in how each hospital gives brachytherapy. Your cancer doctor, radiologist or specialist nurse will explain what will happen.

You usually have antibiotics before brachytherapy and for a few days after to help prevent infection. You may also be given tablets to help you pass urine (pee) more easily after treatment. Your cancer doctor or radiographer can tell you how long to take these for.

Brachytherapy may not be suitable for you if you have a lot of urinary symptoms.

Radioactive seed brachytherapy (LDR)

Radioactive seed brachytherapy is also called low dose rate (LDR) brachytherapy. This is the most common type of brachytherapy for early prostate cancer. You have this treatment done in a day.

You will have a general anaesthetic, or a spinal anaesthetic – this is an injection into the spine. If you have a spinal anaesthetic, you will be awake but will not feel any pain. You will be given a sedative so that you feel sleepy and more relaxed.

Planning radioactive seed brachytherapy

Your doctor passes an ultrasound probe into the anus to the rectum to take pictures of the prostate. This helps them plan how many radioactive seeds are needed and where they should be placed.

Before the scan, you may need to have a special diet for 24 hours. You may also have a small amount of liquid passed into your rectum. This is called an enema. This makes sure your bowel is empty, so the ultrasound picture is clear.

Implanting the seeds

After the planning scan, the doctor passes an ultrasound probe into the anus to the rectum to show the prostate and help guide them. They pass special needles through the skin between the scrotum and anus. This is called the perineum. The doctor inserts the seeds through the needles into the correct position in the prostate.

Sometimes they put a tube (catheter) into the bladder to drain urine during the procedure. You will have the catheter removed before you go home.

After treatment

You can usually go home on the same day, but someone will need to take you. You will not be able to drive because of the anaesthetic. Your [radiotherapy team](#) will explain what to expect after treatment. They will also give you a card to carry with you at all times. It explains the treatment you have had and gives your hospital contact details.

Precautions



The radioactivity from the seeds is absorbed from inside the prostate, so it is safe for you to be around other people. As a precaution, you should avoid periods of close contact with children and anyone pregnant. It is safe for them to be in the same room as you. But you will be asked to keep a certain distance from them. Avoid holding or hugging anyone who is pregnant or children for more than a few minutes each day.

Your [radiotherapy team](#) will give you specific advice about the safety precautions you need to follow and for how long.

Sex



Your [radiotherapy team](#) will advise you to use a condom during sex for the first 2 months after the seeds have been put in. There is a slight risk of passing a seed in your semen when you ejaculate. Place used condoms in a plastic bag and dispose of them in a bin bag.

If you receive anal sex, you will usually be advised to avoid this for several months. This is to protect your partner until most of the radiation has worn off. If you are worried, talk to your radiotherapy team. We have more information on receiving [anal sex after treatment](#).

Your semen may be coloured black or brown for a short time. This is caused by some bleeding during the procedure. It is harmless. You will still need to use contraception to avoid a pregnancy.

Physical activity



After brachytherapy, you need to avoid heavy lifting or energetic physical activity for 2 to 3 days. The area can feel bruised and swollen for a few days. Your GP, cancer doctor or pharmacist can advise you about painkillers.

High dose rate (HDR) brachytherapy

Before HDR brachytherapy you will have a general anaesthetic, or a spinal anaesthetic – this is an injection into the spine. If you have a spinal anaesthetic, you will be awake but will not feel any pain. You will be given a sedative so that you feel sleepy and more relaxed. You may need to stay in hospital overnight.

- For [early prostate cancer](#) – if you are having HDR brachytherapy on its own, you will usually have 2 treatments. If you are having HDR as a boost after external radiotherapy, you only need 1 treatment.
- For [locally advanced prostate cancer](#) – you usually have HDR brachytherapy 1 to 2 weeks before you start external radiotherapy or when you finish external radiotherapy. It is sometimes called a boost dose.

Having the tubes put in

Your doctor passes an ultrasound probe into the anus to the rectum to take pictures of the prostate. This helps them to plan the number of tubes needed and where they should be placed. They may also use a CT scan to help position the tubes.

Your cancer doctor passes thin tubes into your prostate through the skin between the scrotum and back passage (perineum). They may also put a tube into the bladder to drain urine. This is called a catheter. This prevents any swelling of the prostate from stopping you passing urine. They remove the catheter before you go home.

When the tubes are in, you have a [CT](#) or [MRI scan](#). This helps your cancer doctor plan exactly how much radiation to give to the prostate.

Having HDR brachytherapy

You will be taken into a treatment room that has the HDR machine. The tubes are attached to the machine. The machine sends the radioactive source into the tubes in the prostate. A computer monitors how long the radioactive source stays in place. The treatment takes about 1 hour.

The tubes can be uncomfortable, and you must stay lying down while they are in place. After HDR brachytherapy is finished, the radioactive material is returned to the machine and the tubes are removed.

After treatment

After treatment, you have no radioactive material inside you, so there is no risk in being around other people.

Your [radiotherapy team](#) will explain what to expect after treatment. Before you go home, they will give you antibiotics and tablets to help you to pass urine more easily. Take them exactly as explained.

Physical activity

You need to avoid doing heavy lifting or energetic physical activity for 2 to 3 days. The area can feel bruised and swollen for a few days. Your GP, cancer doctor or pharmacist can advise you about [painkillers](#).

Side effects of brachytherapy for prostate cancer

Brachytherapy causes similar side effects to [external beam radiotherapy](#). If you have brachytherapy on its own, side effects may be milder. If you have brachytherapy with external beam radiotherapy, they may be worse. Some side effects may take several weeks to develop and may last for longer.

Erection problems

Brachytherapy for prostate cancer can cause problems getting and keeping an erection. This is called [erectile dysfunction \(ED\)](#).

You may have ED immediately after treatment due to swelling and bleeding. But this can improve as these problems settle down. Sometime ED develops gradually over months or years.

The risk of permanent ED after brachytherapy varies. If you had ED before brachytherapy, it is more likely you will have it afterwards. Having external radiotherapy and hormone therapy will also increase the risk of ED.

Ask your cancer doctor about your risk of ED. If you develop ED, there are different treatments that can help.

After brachytherapy, orgasms might feel different and you might ejaculate little or no semen.

Bowel problems

Brachytherapy may cause some bowel problems for a few weeks after treatment. These are similar to the side effects caused by [external radiotherapy](#). Sometimes they last for months.

Brachytherapy on its own is less likely than external beam radiotherapy to affect the bowel and cause [late bowel effects](#). Late bowel effects are side effects that do not go away after cancer treatment or do not happen until months or years later. If bowel problems do not improve or develop later on, contact your doctor straight away. Always tell them if you notice any bleeding.

Effects on the bladder

These can be more of a problem after brachytherapy. If you already have difficulty passing urine, you will not usually have brachytherapy as it may make [bladder problems](#) worse.

You may see some blood in your urine and semen for a few weeks after treatment. If bleeding gets worse or you notice large blood clots, contact the hospital straight away. Drinking plenty of water helps flush your bladder and prevent blood clots.

You may also:

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- have discomfort or pain passing urine
- need to pass urine urgently and more often.

Drinking plenty of water and avoiding caffeine and alcohol will help improve your symptoms. With HDR brachytherapy, these symptoms should improve after a few weeks. If you had radioactive seed brachytherapy, it may take longer. Your symptoms will improve over the following months as the seeds lose their radioactivity.

Not being able to pass urine

Brachytherapy may cause swelling of the prostate. This may make it difficult to pass urine. If this happens, you may need to have a catheter put in your bladder until the swelling goes down. This usually takes 2 to 4 weeks. Sometimes a catheter needs to be in for a longer time until you can have [surgery to remove part of the prostate \(TURP\)](#).

Sometimes radiotherapy can narrow the urethra. This is called a urethral stricture. This causes difficulty with passing urine. This may happen weeks, months or even years after treatment finishes. There are different treatment options for urethral stricture:

- Sometimes a thin, plastic tube can be passed through the urethra to widen (dilate) it.
- Some people may have a small scope passed into the urethra. Then either a tiny knife or laser is passed through the scope to cut along the stricture to widen the urethra.
- If none of these treatments work, you may need surgery.

Your treatment will depend on factors such as the position and length of the stricture, as well as your age and general health.

Fertility

Having brachytherapy on its own can still cause infertility, but the risk may be lower than with external beam radiotherapy. Infertility means no longer being able to get someone pregnant naturally. If you find this difficult to cope with or are worried about your [fertility](#), talk to your cancer doctor. You may be able to [store sperm](#) before treatment starts.

Benefits and disadvantages of brachytherapy for early prostate cancer

If you have a [low risk](#) cancer and you are not having [active surveillance](#), you may be asked to decide between external beam radiotherapy and brachytherapy. It is important to talk about the possible advantages and disadvantages of each treatment with your healthcare team. You can then decide with them which treatment is right for you.

Benefits

- You usually only need 1 or 2 visits to hospital, rather than weeks of external beam radiotherapy treatment.
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There may be a lower risk of some side effects because a high dose is given to the cancer cells, and a much lower dose is given to the surrounding healthy tissue. For example, you may get fewer bowel side effects with brachytherapy compared to external radiotherapy.

- You can usually get back to your daily routine within a few days.
- Brachytherapy is given internally so it does not affect the skin.

Disadvantages

- You need a general or a spinal anaesthetic.
- With radioactive seeds, you need to avoid close contact with children or anyone who is pregnant for a few months after treatment.
- Short term urinary side effects may be more severe than with external beam radiotherapy.

About our information

This information has been written, revised and edited by Macmillan Cancer Support's Cancer Information Development team. It has been reviewed by expert medical and health professionals and people living with cancer.

References



Below is a sample of the sources used in our prostate cancer information. If you would like more information about the sources we use, please contact us at informationproductionteam@macmillan.org.uk

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We want everyone affected by cancer to feel our information is written for them.

We want our information to be as clear as possible. To do this, we try to:

- use plain English
- explain medical words
- use short sentences
- use illustrations to explain text
- structure the information clearly
- make sure important points are clear.

We use gender-inclusive language and talk to our readers as 'you' so that everyone feels included. Where clinically necessary we use the terms 'men' and 'women' or 'male' and 'female'. For example, we do so when talking about parts of the body or mentioning statistics or research about who is affected.

You can read more about how we produce our information [here](#).

How we can help

Clinical Information Nurse Specialists

Our Cancer Information Nurse Specialists are dedicated cancer nurses available to talk to on our Macmillan Cancer Support Line.



Call us free on 0808 808 0000

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Email the Macmillan Support Line

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Find cancer support services in your area

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